

FIFA World Cup 2026 Simulation Project

Data, Models, Results, and Research State

Generated from the local FIFA_WC_26 workspace on 2026-06-12.

Executive Summary

This project is now a multi-layer FIFA World Cup 2026 simulation and analysis system with primitive, event-based, heuristic, Random Forest, ensemble, calibration, uncertainty, tactical-prior, and external-data-ingestion layers.

- Primary champion consensus begins with France, Brazil, Germany, and Spain.
- The best current validation log loss is from calibrated logistic regression, followed by the weighted ensemble and Random Forest.
- Random Forest explanations point first to squad rating, possession edge, attack-versus-defense, midfield edge, defense-versus-attack, and Elo difference.
- External historical results and World Football Elo ingestion are wired, but the source files are not yet locally present because network access is blocked.

1. Project Objective And Starting Point

The project began as a question: can we build a FIFA World Cup 2026 simulator that is transparent enough to inspect, flexible enough to improve, and rich enough to study why teams become likely winners rather than only outputting a champion list?

The first target was not a perfect prediction engine. The first target was a reproducible baseline: take squads, attach player strength, simulate a tournament thousands of times, store probabilities, and then progressively replace assumptions with better data. That framing matters because every later improvement keeps the earlier baseline available for comparison.

The workspace now contains a complete chain from squad parsing and enrichment to event simulation, machine-learning outcome layers, feature analysis, uncertainty reporting, and external-data ingestion. The report below explains that chain from scratch.

- Tournament scope: the 48-team FIFA World Cup 2026 structure.
- Core outputs: team stage probabilities, champion probabilities, player scoring/assist leaderboards, event totals, feature importances, calibration tables, and data-quality diagnostics.
- Philosophy: keep every assumption auditable and every generated result stored as JSON, CSV, Markdown, LaTeX, or PDF.

2. Raw Data Foundations

The project starts from squad and rating information. The squad file is `guardian_world_cup_2026_player_guide.json`, which represents the active 48-team, 1,248-player working squad dataset. The original baseline also depends on `fifa_squadlists_english.pdf` and `eaafc26_players.csv`, which are large local source files and are intentionally ignored by Git.

The squad rebuild path is handled by `rebuild_squads_from_fifa_pdf.py`. The enrichment path is handled by `enrich_squads_with_eafc.py`. Together, those scripts establish team rosters, player names, official positions, and EAFC/FIFA-style ratings or fallback ratings where no confident EAFC match exists.

The latest squad quality report checks 48 teams and 1248 players. It currently marks the data as `needs_refresh` with 17 squad-shape issues, mostly position-distribution anomalies that should be refreshed against the official PDF before treating player-level outputs as final.

- The primitive baseline uses EAFC-style player ratings as its main strength signal.
- The v2 simulator uses the richer player-performance dataset `player_performance_data_statbunker.json` when building effective ratings and event tendencies.
- The project keeps large generated and source files outside Git where practical, but keeps scripts, templates, and reports in the workspace.

3. Player Performance Data Collection

The second phase tried to move beyond static ratings. Several collectors were created because public football data is fragmented and often blocked for automated access. `collect_player_performance_data.py` targets SofaScore-style profile evidence; `collect_player_performance_data_edge.py` tries browser-assisted access when direct requests are blocked; `collect_player_performance_data_providers.py` supports key-based providers such as Sportmonks, API-Football/API-Sports, and FootyStats; `collect_player_performance_data_statbunker.py` collects public StatBunker competition tables.

StatBunker became the current free fallback. It gives useful public player-table data such as appearances, goals, assists, cards, penalties, fantasy points where requested, and goalkeeper clean-sheet data. It does not fully solve the problem because it lacks many event-level metrics such as xG, xA, pressures, carries, tackles, saves, and SofaScore-style match ratings.

Because player coverage is uneven, the research pipeline now creates `player_data_coverage.csv`. This explicitly identifies teams where low-confidence players or zero-minute players weaken the model. That turns data quality into a measurable object rather than a vague concern.

- SofaScore access was unreliable because direct endpoints often return 403 outside a browser session.
- Browser-assisted collection can recover some profile evidence, but is not a robust full dataset.
- Paid/key-based providers remain the best future route for complete player-level event data.
- Current lowest-confidence coverage includes Panama, Senegal, Mexico, Scotland, Canada, Ivory Coast, Czech Republic, and the United States.

4. Primitive Baseline Simulator

The first working model is `simulate_world_cup.py`. It exists as an intentionally simple baseline. Each tournament epoch simulates the group stage, ranks group tables, advances top-two teams plus the eight best third-place teams, simulates knockouts, simulates the third-place match, and stores team and player outputs.

Team strength in this baseline is built from best XI average, top-18 depth, and full-squad average. Match goals are generated from a Poisson model. Knockout draws are resolved through extra time and penalties. Player goals and assists are assigned probabilistically using player position and rating.

The primitive 10,000-run output is `simulation_results_10000.json`, with a visual report in `simulation_report_10000`. It remains valuable because it tells us what the simplest rating-and-goals model thinks before later event, Elo, and ML layers alter the result.

- Primitive champion leader: France at 12.50%.
- Other high primitive probabilities include Spain, England, Brazil, Portugal, Germany, Argentina, and the Netherlands.
- This model does not include possession, fatigue, cards, travel, climate, Random Forest adjustment, player availability, or calibrated historical supervision.

5. V2 Event Simulator

`simulate_world_cup_v2.py` is the major modelling upgrade. It keeps the tournament structure but changes the match engine from a simple Poisson-only model into a compressed event simulator. A 90-minute match is represented by 9 simulated minutes, and extra time is represented by 3 simulated minutes.

The v2 engine adds possession, fouls, yellow cards, red cards, second yellows, player removals, substitutions, fatigue, set-piece goals, game-state behavior, extra time, penalties, rest-day context, date-aware scheduling, host-country signals, venue/climate/travel priors, and constrained Round-of-32 routing.

Only the team with possession can score during an event tick. Cards and fatigue alter team state. Red cards and second yellows remove players. Substitutes are modelled as fresher than tired starters. These mechanics are still simplified, but they are much closer to the match process we want to study than the primitive baseline.

- The engine exports pandas-compatible feature frames for inspection.
- It can run CPU-parallel tournament batches, which is why 10,000-run v2 simulations were practical on the Ryzen 7 8845HS.
- It supports a heuristic internal forest or scikit-learn Random Forest match outcome layer.

6. National-Team Elo And Rating Priors

The simulator now reads `national_team_elo_ratings.json` as a national-team prior. The file can contain all national teams, not just the World Cup teams. The simulator consumes the teams in the tournament structure, while aliases help historical match training match names such as USA, Czechia, Korea Republic, and Cote d'Ivoire.

The current import report contains 177 teams from raw World Football Elo as of 2026-06-11, sourced as World Football Elo Ratings System user-supplied export. Missing required World Cup teams: 0.

The active ratings now come from `external_data/world_football_elo/international_football_elo_20260611.json`. The earlier FIFA-rank-derived fallback was preserved in `external_data/world_football_elo` for audit and rollback.

- Preferred priority: `world_football_elo` first, then `elo/rating`, then FIFA points, then FIFA rank-derived Elo, then squad proxy fallback.

- World Football Elo source is configured as <https://www.eloratings.net/>.
- A template exists at `world_football_elo_current_template.csv`.

7. Historical Match Data Path

The Random Forest and research pipeline are prepared for supervised historical match training, but the local workspace does not yet contain the historical results CSV. The chosen source is `martj42/international_results`, specifically its `results.csv` with `date`, `home_team`, `away_team`, `home_score`, `away_score`, `tournament`, `city`, `country`, and `neutral` columns.

`fetch_external_football_data.py` can download and normalize that CSV when network access is available. In this workspace, direct access to `raw.githubusercontent.com` is blocked, so the manifest records the failed download and tells us where the raw file should be placed: `external_data/international_results/results.csv`.

The current research pipeline therefore reports historical data readiness as missing, with 0 usable historical matches. As a result, model validation currently measures the simulator's synthetic/profile calibration, not real-world historical predictive performance.

- Once `results.csv` is placed locally, `fetch_external_football_data.py` normalizes it into `historical_international_results.csv`.
- `simulate_world_cup_v2.py` and `analyze_feature_importance.py` already accept `--historical-matches`.
- If the historical file is small, the project blends it with synthetic calibration rows instead of discarding it.

8. Random Forest And Model Training Layer

The machine-learning layer was introduced so we can study factors rather than only simulate outcomes. The Random Forest consumes match-level feature differences: Elo difference, squad rating difference, attack-versus-defense, defense-versus-attack, midfield difference, goalkeeper difference, possession difference, discipline difference, data confidence, host context, rest, fatigue, lineup replacements, altitude, travel, heat, humidity, venue altitude, stage type, draw allowance, and match-day index.

In the current feature-importance run, the training source is `synthetic_profile_calibration` with 6768 rows. These are synthetic profile-calibration rows because historical match data is not yet present.

The best current validation model is `calibrated_logistic_sigmoid` with log loss 0.75862, accuracy 0.69208, and expected calibration error 0.05681.

- Random Forest is used as one model, not the only model.
- The research layer also trains Extra Trees, histogram gradient boosting, logistic regression, and calibrated logistic regression.
- The ensemble weights are derived from validation log loss so stronger validation models receive slightly higher influence.

9. 10,000-Run Simulation Comparisons

The project now stores comparable 10,000-run outputs for three primary model families: primitive EAFC Poisson, v2 heuristic internal forest with NT Elo, and v2 scikit-learn Random Forest with NT Elo. It also preserves a legacy v2 Random Forest run that used squad-proxy Elo before the NT Elo import.

The leading consensus teams across the primary 10,000-run models are France (17.02%), Brazil (15.91%), Germany (15.39%), Spain (15.18%).

`generate_10000_model_comparison_report.py` turns those runs into a common table structure. That matters because each raw simulation JSON has a slightly different schema, and the comparison report provides a single long team summary, champion matrix, metadata file, and player leaderboards.

- Primitive EAFC Poisson top winner: France.
- V2 heuristic with NT Elo top winner: Brazil.
- V2 Random Forest with NT Elo top winner: Brazil.
- The v2 event models concentrate more probability among Brazil, Germany, France, and Spain than the primitive model does.

10. Feature Importance And Explainability

`analyze_feature_importance.py` was added to answer a more important research question: which factors actually drive the model's predictions? It trains several comparison models, exports Random Forest trees, calculates built-in tree importances, permutation importances, logistic coefficients, local occlusion effects, partial dependence, ICE curves, validation group ablations, and tournament simulation ablation scenarios.

The strongest Random Forest permutation features are `squad_rating_diff`, `possession_diff`, `attack_vs_defense`, `midfield_diff`, `defense_vs_attack`, `elo_diff`, and `gk_diff`. Local occlusion tells a similar story: neutralizing squad rating, possession, attacking matchup, midfield, and defensive matchup moves the model's true-class probabilities the most.

At the group level, `unit_matchups` is the most important ablation group. This group includes `attack_vs_defense`, `defense_vs_attack`, `midfield_diff`, `gk_diff`, and `possession_diff`. In plain English, the model is not only looking at which team is stronger overall; it cares about how specific units match up.

- Feature importance outputs live in feature_importance_results_seed_20260618.
- Random Forest trees were exported to text and JSON for inspection.
- The feature report warns that current importances explain simulator assumptions until real historical rows are supplied.

11. Calibration, Uncertainty, And Ensemble Research Layer

`run_research_upgrade_pipeline.py` adds a research layer on top of the simulator outputs. It trains comparison models, evaluates accuracy, balanced accuracy, log loss, multiclass Brier score, and expected calibration error, then writes calibration bins and ensemble weights.

It also computes Wilson 95 percent uncertainty intervals for every stored 10,000-run stage probability. This changes how probabilities should be read: Brazil at about 20 percent is not a single exact truth; it is an estimate with simulation uncertainty and model uncertainty.

The pipeline also writes tactical priors and player data coverage diagnostics. Tactical priors are inferred from current squad profiles, so they are useful modelling features but should not be mistaken for validated event-style data.

- Current best validation log loss: `calibrated_logistic_sigmoid`.
- Current strongest calibration by ECE: histogram gradient boosting in the stored research report.
- Consensus ensemble and uncertainty tables live in `research_upgrade_results_seed_20260622`.

12. How To Read The Results

The results should be read as model-conditioned tournament probabilities. They are not declarations of what will happen. Every number depends on the current squad data, current player-performance coverage, current Elo prior, current fixture assumptions, and current match engine.

The most useful result is not just the top champion. The useful result is how different model families disagree. For example, the primitive baseline gives France the highest title chance, while both v2 event models make Brazil the top title candidate. That disagreement tells us that event-engine assumptions and rating priors materially affect the output.

The player leaderboards are especially sensitive to squad-shape and player-data quality. They are good for checking whether the simulator behaves plausibly, but they should not be shared as final player award predictions until the squad quality issues and player coverage weaknesses are resolved.

- Use consensus tables for a conservative model-family view.
- Use uncertainty intervals to avoid over-reading small percentage differences.
- Use feature importance and ablation tables to understand model behavior.
- Use coverage tables to decide what data to improve next.

13. Reproducible Build Ledger

This section records what was actually built in the workspace. It is included so the report can be shared with someone who has not watched the project evolve conversation by conversation.

The project did not jump straight to a Random Forest. It moved through a deliberate sequence: establish a squad dataset, build a primitive baseline, enrich player evidence, add a more realistic event engine, add rating priors, add machine-learning layers, run comparable 10,000 simulations, explain the model, and finally add research diagnostics and external-data ingestion.

- Created or maintained `rebuild_squads_from_fifa_pdf.py` to parse FIFA squad-list inputs into the working squad JSON.
- Created or maintained `enrich_squads_with_eafc.py` to attach EAFC/FIFA-style player ratings, detailed positions, and fallback ratings.

- Ran the primitive simulator `simulate_world_cup.py` and generated `simulation_results_10000.json` plus `simulation_report_10000`.
- Created `generate_simulation_report.py` for the original HTML/SVG simulation report.
- Built performance data collectors for SofaScore-style pages, browser-assisted fallback, key-based providers, and StatBunker public tables.
- Stored `player_performance_data_statbunker.json` as the current free public performance-data layer.
- Created `simulate_world_cup_v2.py` with possession, fouls, cards, removals, fatigue, substitutions, set pieces, compressed match time, extra time, penalties, rest, venue, climate, travel, and bracket routing.
- Added `--trace-one` output so one simulated tournament can be watched and recorded in the console.
- Added CPU-parallel v2 tournament execution for large runs on the Ryzen 7 8845HS.
- Created `import_national_team_elo.py` and `national_team_elo_ratings.json` to support all-national-team rating priors and aliases.
- Created `world_cup_2026_venue_context.json` for group-level and knockout-level venue, climate, altitude, and travel priors.
- Added scikit-learn Random Forest support with synthetic calibration and optional historical CSV training.
- Exported Random Forest trees to readable text and JSON using `export_random_forest_trees.py`.
- Ran 10,000 simulations for primitive EAFC Poisson, v2 heuristic with NT Elo, and v2 Random Forest with NT Elo.
- Created `generate_10000_model_comparison_report.py` to compare all stored 10,000-run model outputs.
- Created `analyze_feature_importance.py` to compute model metrics, feature importance, local occlusion, PDP/ICE, and ablation studies.
- Created `run_research_upgrade_pipeline.py` for calibration, ensemble weights, uncertainty intervals, tactical priors, player-data coverage, source manifests, and a local dashboard.
- Created `fetch_external_football_data.py` to normalize `martj42/international_results` and import World Football Elo CSV/JSON files.
- Created `generate_shareable_latex_report.py` to produce this PDF and matching LaTeX source from the stored outputs.

14. What The Stored Files Mean

The project stores raw model outputs separately from reports. This is intentional: reports can be regenerated, while CSV and JSON outputs remain the auditable data behind those reports.

The most important folder for current study is `research_upgrade_results_seed_20260622`. It contains validation metrics, calibration bins, uncertainty intervals, consensus champion probabilities, tactical priors, player coverage, and the local HTML dashboard. The most important folder for model comparison is `simulation_10000_model_comparison`. The most important folder for feature interpretation is `feature_importance_results_seed_20260618`.

- `simulation_10000_model_team_summary.csv` is the long table of team stage probabilities by model.
- `simulation_10000_model_champion_matrix.csv` is the wide title-probability comparison across models.
- `simulation_consensus_ensemble.csv` is the current primary cross-model champion consensus.
- `simulation_stage_uncertainty.csv` gives Wilson intervals for all stage probabilities.
- `model_validation_metrics.csv` compares candidate predictive models on accuracy, balanced accuracy, log loss, Brier score, and calibration error.
- `feature_importance_permutation.csv` gives model-agnostic feature importance by shuffling each feature.

- feature_group_ablation_validation.csv estimates how much validation performance drops when a feature group is neutralized.
- player_data_coverage.csv identifies which teams most need better player-level data before player outputs should be trusted.
- external_football_data_manifest.json records the configured historical-results and World Football Elo sources.

Model Inventory

model	epochs	seed	elo_source	forest	training_rows
Primitive EAFC Poisson	10000	20260605		none	
V2 Heuristic Internal Forest + NT Elo	10000	20260620	FIFA/Coca-Cola Men's World Ranking ranks converted to Elo-scale prior	heuristic_internal_forest	0.0
V2 sklearn Random Forest + NT Elo	10000	20260621	FIFA/Coca-Cola Men's World Ranking ranks converted to Elo-scale prior	sklearn_random_forest	6768.0
V2 sklearn Random Forest + squad-proxy Elo legacy	10000	20260618	squad_proxy	sklearn_random_forest	6768.0

10,000 Simulation Results

Cross-Model Champion Consensus

team	mean title %	median title %	range	v2 mean %
France	17.02%	19.09%	6.96%	19.27%
Brazil	15.91%	19.23%	11.67%	19.66%
Germany	15.39%	18.30%	11.64%	19.02%
Spain	15.18%	17.11%	7.92%	17.64%
Argentina	7.71%	7.67%	1.59%	7.72%
England	6.82%	6.76%	3.39%	5.96%
Portugal	5.70%	4.36%	4.04%	4.36%
Netherlands	4.62%	4.30%	2.79%	3.84%
Belgium	2.32%	1.39%	3.37%	1.25%
Turkey	0.89%	0.30%	1.88%	0.28%
Croatia	0.88%	0.06%	2.49%	0.06%
Switzerland	0.65%	0.24%	1.38%	0.20%
Morocco	0.65%	0.15%	1.64%	0.11%
Norway	0.61%	0.13%	1.48%	0.12%

Champion Uncertainty

model	team	title %	95% low	95% high
Primitive EAFC Poisson	France	12.50%	11.87%	13.16%
Primitive EAFC Poisson	Spain	10.26%	9.68%	10.87%
Primitive EAFC Poisson	England	8.55%	8.02%	9.11%
Primitive EAFC Poisson	Brazil	8.42%	7.89%	8.98%

model	team	title %	95% low	95% high
Primitive EAFC Poisson	Portugal	8.39%	7.86%	8.95%
V2 Heuristic Internal Forest + NT Elo	Brazil	19.23%	18.47%	20.01%
V2 Heuristic Internal Forest + NT Elo	France	19.09%	18.33%	19.87%
V2 Heuristic Internal Forest + NT Elo	Germany	18.30%	17.55%	19.07%
V2 Heuristic Internal Forest + NT Elo	Spain	18.18%	17.44%	18.95%
V2 Heuristic Internal Forest + NT Elo	Argentina	8.52%	7.99%	9.08%
V2 sklearn Random Forest + NT Elo	Brazil	20.09%	19.32%	20.89%
V2 sklearn Random Forest + NT Elo	Germany	19.75%	18.98%	20.54%
V2 sklearn Random Forest + NT Elo	France	19.46%	18.70%	20.25%
V2 sklearn Random Forest + NT Elo	Spain	17.11%	16.38%	17.86%
V2 sklearn Random Forest + NT Elo	Argentina	6.93%	6.45%	7.44%

Validation, Calibration, and Ensemble

model	accuracy	balanced acc.	log_loss	Brier	ece
calibrated_logistic_sigmoid	0.692	0.550	0.759	0.434	0.057
weighted_validation_ensemble	0.690	0.572	0.773	0.443	0.076
random_forest	0.671	0.589	0.794	0.458	0.086
hist_gradient_boosting	0.689	0.557	0.802	0.450	0.044
extra_trees	0.659	0.601	0.819	0.477	0.111
logistic_l2	0.624	0.580	0.828	0.485	0.097

Ensemble Weights

model	log_loss	weight
calibrated_logistic_sigmoid	0.759	0.211
random_forest	0.794	0.201
hist_gradient_boosting	0.802	0.199
extra_trees	0.819	0.195
logistic_l2	0.828	0.193

Feature Importance

feature	perm. delta	std
squad_rating_diff	0.059	0.003
possession_diff	0.033	0.002
attack_vs_defense	0.027	0.002
midfield_diff	0.016	0.001
defense_vs_attack	0.015	0.001
elo_diff	0.011	0.002
gk_diff	0.005	0.001
data_confidence_diff	0.002	0.001
travel_diff	0.002	0.001
match_day_index	0.001	0.001

Feature Group Ablation

group	loss increase	acc. drop
unit_matchups	0.023	0.012

group	loss increase	acc. drop
team_strength	0.005	0.005
rest_fatigue_availability	0.005	0.000
discipline	0.003	-0.002
venue_climate_travel	0.002	-0.001
tournament_structure	0.001	-0.003
host_context	0.000	-0.002
data_quality	0.000	0.005

Player and Tactical Layer

team	pressing	counter	low block	set pieces	tempo	confidence
Spain	88.510	86.660	86.779	88.081	88.843	0.937
England	88.022	87.362	85.099	88.077	88.626	0.952
Germany	87.907	85.891	85.879	87.350	88.229	0.873
Portugal	87.415	86.117	85.068	87.154	87.822	0.859
France	87.071	88.961	87.706	88.144	87.991	0.902
Brazil	86.695	86.388	87.998	86.779	87.329	0.860
Argentina	86.344	86.809	85.042	87.022	87.363	0.831
Netherlands	86.038	85.180	83.099	85.962	86.477	0.920
Belgium	85.534	84.431	85.303	85.234	85.911	0.781
Norway	82.846	82.057	78.064	82.963	83.653	0.781
Croatia	82.673	81.862	80.509	82.843	83.463	0.692
Turkey	82.250	80.674	79.729	82.064	82.872	0.696

Player Data Coverage Weak Spots

team	avg confidence	low conf.	zero min.	primary_sources
Panama	0.501	18	24	statbunker:18; wikipedia:4; transfermarkt:4
Senegal	0.585	14	25	statbunker:24; wikipedia:1; transfermarkt:1
Mexico	0.633	11	17	statbunker:15; transfermarkt:10; wikipedia:1
Scotland	0.636	11	24	statbunker:23; transfermarkt:2; wikipedia:1
Canada	0.644	11	21	statbunker:17; transfermarkt:7; wikipedia:2
Ivory Coast	0.645	10	22	statbunker:19; transfermarkt:6; wikipedia:1
Czech Republic	0.649	11	22	statbunker:19; transfermarkt:5; wikipedia:2
United States	0.659	10	23	statbunker:19; transfermarkt:5; wikipedia:2
Australia	0.669	11	14	transfermarkt:12; statbunker:11; wikipedia:3
Qatar	0.670	12	22	statbunker:12; wikipedia:8; transfermarkt:4; manual_web_research:2

Player Leaderboards

model	rank	player	team	goals/tourn.
Primitive EAFC Poisson	1	Lautaro Martinez	Argentina	1.449
Primitive EAFC Poisson	2	Kylian Mbappe	France	1.446
Primitive EAFC Poisson	3	Romelu Lukaku	Belgium	1.398
Primitive EAFC Poisson	4	Cristiano Ronaldo	Portugal	1.391
Primitive EAFC Poisson	5	Julian Alvarez	Argentina	1.385
V2 Heuristic Internal Forest + NT Elo	1	Kylian Mbappe	France	2.946
V2 Heuristic Internal Forest + NT Elo	2	Lautaro Martinez	Argentina	2.739
V2 Heuristic Internal Forest + NT Elo	3	Cristiano Ronaldo	Portugal	2.719
V2 Heuristic Internal Forest + NT Elo	4	Harry Kane	England	2.629
V2 Heuristic Internal Forest + NT Elo	5	Romelu Lukaku	Belgium	2.540
V2 sklearn Random Forest + NT Elo	1	Kylian Mbappe	France	2.904
V2 sklearn Random Forest + NT Elo	2	Cristiano Ronaldo	Portugal	2.720
V2 sklearn Random Forest + NT Elo	3	Harry Kane	England	2.679
V2 sklearn Random Forest + NT Elo	4	Lautaro Martinez	Argentina	2.593
V2 sklearn Random Forest + NT Elo	5	Romelu Lukaku	Belgium	2.491

Top Assisters

model	rank	player	team	assists/tourn.
Primitive EAFC Poisson	1	Jude Bellingham	England	0.828
Primitive EAFC Poisson	2	Florian Wirtz	Germany	0.792
Primitive EAFC Poisson	3	Jamal Musiala	Germany	0.757
Primitive EAFC Poisson	4	Matheus Cunha	Brazil	0.754
Primitive EAFC Poisson	5	Raphinha	Brazil	0.751
V2 Heuristic Internal Forest + NT Elo	1	Florian Wirtz	Germany	1.814
V2 Heuristic Internal Forest + NT Elo	2	Jamal Musiala	Germany	1.696
V2 Heuristic Internal Forest + NT Elo	3	Lamine Yamal	Spain	1.645
V2 Heuristic Internal Forest + NT Elo	4	Raphinha	Brazil	1.636
V2 Heuristic Internal Forest + NT Elo	5	Jude Bellingham	England	1.603
V2 sklearn Random Forest + NT Elo	1	Florian Wirtz	Germany	1.816
V2 sklearn Random Forest + NT Elo	2	Jamal Musiala	Germany	1.694
V2 sklearn Random Forest + NT Elo	3	Raphinha	Brazil	1.652
V2 sklearn Random Forest + NT Elo	4	Lamine Yamal	Spain	1.625

model	rank	player	team	assists/tourn.
V2 sklearn Random Forest + NT Elo	5	Jude Bellingham	England	1.617

External Data and Elo

Historical data readiness: missing. Usable historical matches detected: 0. The current import report contains 177 teams from raw World Football Elo as of 2026-06-11, sourced as World Football Elo Ratings System user-supplied export. Missing required World Cup teams: 0.

name	purpose	local_filename	status
International football results	Historical match outcomes for supervised model training and calibration	external_data/international_results/results.csv or historical_international_results.csv	download_blocked_in_workspace; place the CSV locally to activate historical training
World Football Elo Ratings	Time-aware national-team Elo before each match and current NT priors	external_data/world_football_elo/international_football_elo_20260611.json; national_team_elo_snapshots.csv later	baseline_imported_2026-06-11; historical time-aware Elo snapshots still needed; rerun simulations for raw-Elo result artifacts
Current project squad/player data	Team strength, unit matchups, discipline priors, and tactical priors	guardian_world_cup_2026_player_guide.json; player_performance_data_statbunker.json	available

Caveats and Next Inputs

- Place martj42 results.csv at external_data/international_results/results.csv, then run fetch_external_football_data.py.
- Use the imported 2026-06-11 World Football Elo snapshot as the current tournament baseline; add time-aware Elo snapshots later if historical backtesting needs match-date priors.
- Rerun the 10,000-simulation model comparison now that raw Elo is active.
- Rerun the research pipeline after those files exist so the model can move from synthetic calibration toward historical supervision.